

Prasun Rimal

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EDUCATION

St. Joseph's University, New York

B.S. in Mathematics/Computer Science & B.S. in Mathematics

May 2026

Brooklyn, NY

GPA: 3.811/4.000

Honors: Sigma Iota Chi, Delta Epsilon Sigma, Dean's List: 6 semesters

Selected Coursework: Data Structures, Algorithm Design and Analysis, Software Engineering, Database Systems, Operating Systems, Advanced Computer Programming, Data Science, Probability and Statistics, Linear Algebra

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, R, C++, MATLAB, HTML, CSS

Backend: Spring Boot, Spring Security, Spring Data JPA, JWT, Flask, Django, Express, REST APIs, Microservices

Frontend: React, Next.js, Vite, Tailwind CSS, Recharts, Leaflet, JavaFX

AI/ML: LangChain, Prompt Engineering, NLP, Pandas, NumPy, R/tidyverse, ggplot2, Logistic regression, Data visualization

Databases/Cloud: PostgreSQL, MySQL, MongoDB, Oracle, Neon, Render, Vercel, Docker, Kubernetes, AWS

Core Concepts: Data structures, Algorithms, Graph algorithms, Optimization, OOP, Design patterns, Agile development

PROJECTS

MockMarket: Full-Stack Paper Trading Platform

[Live Demo](#) — [GitHub](#)

Java 21, Spring Boot, React, TypeScript, PostgreSQL, JWT, Finnhub API, Vercel, Render, Neon

- Built and deployed a production-ready, full-stack paper trading platform giving each registered user a simulated \$100,000 account for buying, selling, and tracking stocks with live market quotes.
- Architected the backend as independently deployable microservices—auth, trading engine, market-data integration, and portfolio analytics—following service-oriented design principles with protected REST APIs.
- Implemented precise financial calculations with **BigDecimal**, including cash balance updates, average cost tracking, holdings value, portfolio value, and gain/loss calculations.
- Configured multi-environment deployment across Vercel (frontend), Render (backend), and Neon (cloud PostgreSQL) with environment-based secrets management and production/development separation.
- Integrated Finnhub market data for live quotes and company search; designed a polished React/TypeScript dashboard with portfolio analytics, trade history, watchlist management, and responsive trading-terminal styling.

Manhattan Last-Mile Delivery Optimizer: SURF Research Project

[GitHub](#)

Python, Flask, NetworkX, Pandas, Leaflet, OpenStreetMap

- Built the technical artifact for a Summer Undergraduate Research Fellowship project focused on last-mile delivery optimization in Manhattan; scoped requirements and iterated on design through weekly advisor reviews.
- Implemented Dijkstra's algorithm and A* search to compare shortest-distance routes against delivery-aware routes in a graph model enriched with traffic, curb-access, parking/loading, and truck-route constraints.
- Designed a multi-objective cost function that balances distance, congestion, curb limitations, and route legality to better reflect real urban delivery conditions.
- Developed an interactive Flask/Leaflet web app that visualizes baseline and optimized routes; demonstrated an optimized sample route that added only 35.01 meters while reducing modeled delivery cost by 34.62 units.

EXPERIENCE

Summer Research Fellow, Summer Undergraduate Research Fellowship (SURF)

May 2025 – August 2025

St. Joseph's University, New York

Brooklyn, NY

- Conducted faculty-advised undergraduate research on graph-based optimization strategies for last-mile delivery under real-world urban constraints, gathering and scoping project requirements through structured advisor sessions.
- Iterated on algorithm design and modeling methodology through an agile weekly review cycle, flagging technical risks and refining the approach based on advisor feedback.
- Wrote a formal research paper documenting problem motivation, data sources, algorithmic approach, optimization framework, findings, and limitations; delivered findings presentation to a stakeholder audience.

LEADERSHIP

Vice President, South Asian Student Association

August 2025 – May 2026

St. Joseph's University, New York

Brooklyn, NY

- Helped lead a campus cultural organization focused on community building, student representation, and South Asian cultural programming.
- Coordinated with executive board members to plan events, organize meetings, promote student engagement, and support inclusive programming for the broader campus community.
- Contributed to event logistics, outreach, and communication efforts for cultural celebrations, student activities, and organization-led initiatives.

AWARDS AND HONORS

- Sigma Iota Chi Honor Society
- Delta Epsilon Sigma Honor Society
- Dean's Scholarship Recipient
- Dean's List, 6 semesters
- Summer Undergraduate Research Fellowship (SURF) Recipient